

Project 1

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Abstract

In this project, we study the solution to the ordinary differential equation (ODE) given by a damped harmonic oscillator. We implement an ODE solver using Euler's method and 4th order Runge-Kutta method in Python. Additionally, we solve for the work done by the damping force by implementing an integration solver using the simple Riemann sum, the trapezoidal rule, and Simpson's rule. We compare the accuracy of each method against the implementation provided by the SciPy library. Finally, we will validate our results by confirming the work energy theorem and comparing the solutions for overdamped and underdamped situations.

Introduction

Consider an oscillator, such as a spring, subject to a displacement force given by Hook's Law

$$F_h = -kx$$

where k is the spring constant and x is the displacement. This force alone would give rise to simple harmonic motion. However, if a friction force, with damping coefficient c , were added to the system

$$F_f = -c\dot{x}$$

the motion would no longer be simple harmonic. Instead, the motion would be given by the solution to the equation of motion

$$m\ddot{x} + c\dot{x} + kx = 0.$$

The work done by the friction force is given by

$$W = \int F_f \cdot dx = \int -c\dot{x} \frac{dx}{dt} dt = \int -c\dot{x}^2 dt.$$

In this paper we will study the solution and properties of the damped harmonic oscillator and the work done by the friction force.

ODE Solver

There are two methods we will implement in order to solve the equation of motion given above, Euler's Method and the 4th order Runge-Kutta method. We then will compare our implementation of these methods to SciPy's.

Euler's Method

Let

$$\frac{S(t)}{dt} = F(t, S(t))$$

be a first order ordinary differential equation (ODE) where F is some function that returns the change of state and t are the values over an interval $[t_o, t_f]$ with spacing h . Euler's method approximates the solution at each step $t_j + 1$ using the solution to the previous step, $S(t_j)$. This approximation is given by

$$S(t_{j+1}) = S(t_j) + hF(t_j, S(t_j)).$$

If we know the initial condition $S(t_0)$, then we can approximate S at any step.

4th Order Runge-Kutta Method

Euler's Method is a linear approximation which comes from the first two terms of the Taylor series. The Runge-Kutta methods keep higher terms in the Taylor series to increase the accuracy of the approximate solution. In particular, the 4th order Runge-Kutta method uses the weighted

average of four points derived from the higher order terms in the Taylor series

$$k_1 = F(t_j, S(t_j))$$

$$k_2 = F(t_j + \frac{h}{2}, S(t_j) + \frac{1}{2}k_1h)$$

$$k_3 = F(t_j + \frac{h}{2}, S(t_j) + \frac{1}{2}k_2h)$$

$$k_4 = F(t_j + h, S(t_j) + k_3h).$$

Then the approximate solution is

$$S(t_{j+1}) = S(t_j) + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

Both of these methods provide solutions to first order ODE's. However, our equation of motion is a second order ODE. Thus when implementing these methods we must define a system of two first order ODE's

$$\begin{cases} \dot{x} = y \\ \dot{y} = -(\frac{c}{m})y - (\frac{k}{m})x. \end{cases}$$

Integration Solver

We used 3 different methods to solve for the work done by the friction force.

Riemann Sum

The Riemann sum is calculated by approximating that the area under the integrand curve is the sum of areas of rectangles on sub-intervals of $[t_o, t_f]$. This looks like

$$\int_{t_o}^{t_f} f(t)dt \approx \sum h f(t_i)$$

where h is the width of each rectangle and $f(t_i)$ is the height.

Trapezoidal Rule

Instead of using rectangles, the trapezoidal rule uses trapezoids to better fit the area under the curve. The trapezoid may look different for each sub-interval. This approximation is of the form

$$\int_{t_o}^{t_f} f(t)dt \approx \sum h \frac{f(t_i) + f(t_{i+1})}{2}.$$

Simpson's Rule

Simpson's Rule fits a quadratic polynomial through every three consecutive data points and integrates the polynomial exactly. This approximation is given by

$$\int_{t_o}^{t_n} f(t) dt \approx \frac{h}{3} \left[f(t_0) + f(t_n) + 4 \sum_{\substack{i=1 \\ i \text{ odd}}}^{n-1} f(t_i) + 2 \sum_{\substack{i=2 \\ i \text{ even}}}^{n-2} f(t_i) \right].$$

Simpson’s Rule requires an even number of intervals since each application of the method uses two sub-intervals spanned by three points. Additionally, though we have assumed uniform spacing so far, for Simpson’s Rule it is specifically required.

Results

When using a reasonable damping coefficient ($c = 0.1$, with mass and spring constant both equal to 1.0) we find the oscillatory solution shown in Figure 1 as expected. We see that the RK4 method is more accurate to the solution given by SciPy’s solver than Euler’s method. This is also as expected since the RK4 uses more terms in the Taylor expansion to approximate a solution.

Additionally, the work calculated is negative as expected as the damping force removes kinetic energy from the system. The 3 different integral solver methods were very consistent and just as accurate as SciPy’s versions.

Method	Value
Riemann Sum	-0.429148
Trapezoidal	-0.429424
Simpson’s	-0.429422
SciPy Trapezoidal	-0.429424
SciPy Simpson’s	-0.429422
ΔE	-0.429422

Table 1: Comparison of numerical integration methods and ΔE .

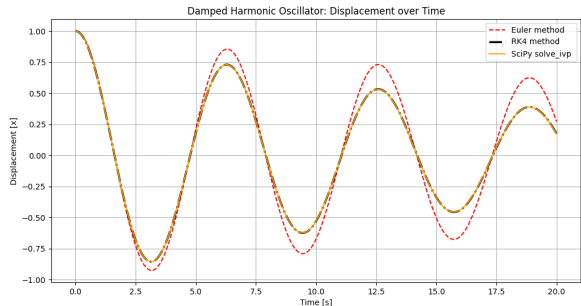


Figure 1: Solutions to the damped oscillator ODE using 3 different ODE solvers

Damping Effects

To physically validate the results we will compare the solutions for an overdamped ($c = 10.0$), underdamped ($c = 0.001$) and critically damped ($c = 2.0$) situation. As shown in the figures below, the solutions behave just as expected for each case. The overdamped case has no oscillatory motion while the underdamped case continues to oscillate farther. The critically damped case passes equilibrium once then motion is terminated. These solutions are consistent with physical expectations.

Work Energy Theorem

To further validate our calculations, we check that the work-energy theorem holds. This theorem states that the total work done is equal to the change in kinetic energy. We calculate the work done by the damping force with our integral solvers. Then we know the work done by the displacement force is given by the change in potential energy. Thus we are checking

$$W_{tot} = W_f + W_h = \Delta KE$$

$$W_f = \Delta(KE + U)$$

We found that this holds for each damping coefficient tested as outputted in the code. This is demonstrated by the data shown in Table 1.

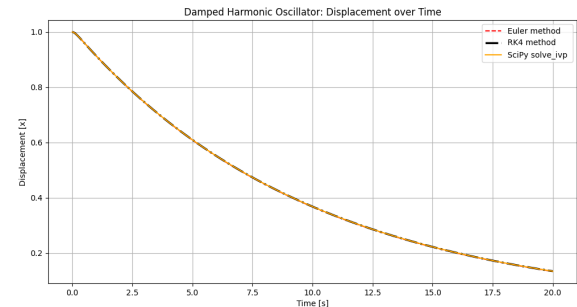


Figure 2: Solutions to the overdamped oscillator ODE using 3 different ODE solvers

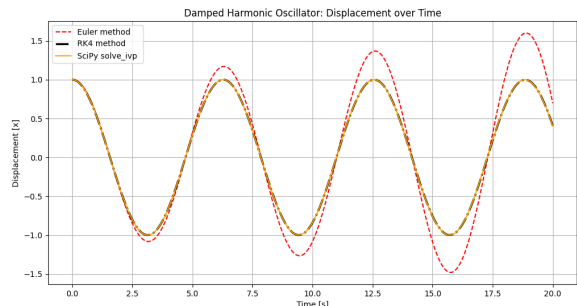


Figure 3: Solutions to the underdamped oscillator ODE using 3 different ODE solvers

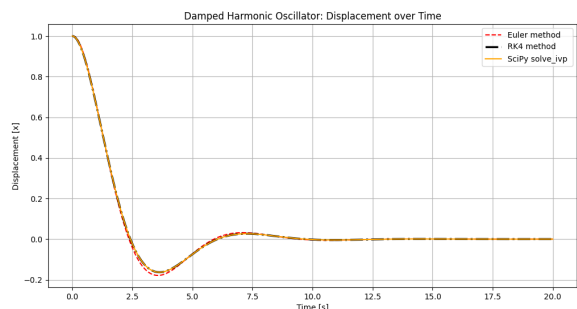


Figure 4: Solutions to the critically damped oscillator ODE using 3 different ODE solvers

Conclusion

In this project, we successfully implemented numerical methods to solve the damped harmonic oscillator and to compute the work done by the damping force. By comparing Euler’s method and the 4th order Runge–Kutta method with SciPy’s ODE solver, we confirmed that RK4 provides improved accuracy as expected. Our integration methods Riemann sum, trapezoidal rule, and Simpson’s rule each produced results in agreement with SciPy’s implementations, demonstrating both the consistency and reliability of our solvers.

The physical validation from examining underdamped, overdamped, and critically damped cases shows that our results match theoretical expectations, reinforcing the accuracy of our solutions. Finally, by verifying the work energy theorem, we confirmed that the computed work done is consistent with the corresponding changes in energy which further validates our solutions.

References

[1] Qingkai Kong, Timmy Siau, and Alexandre Bayen. *Python Programming and Numerical Methods: A Guide for Engineers and Scientists*. Elsevier, Amster-

dam; Boston, 1 edition, 2020. First edition, published December 2 2020.